Corpus Text Analysis Tools

Contents

The NLP Suite provides several corpus-level text analysis tools accessible from the Corpus Statistics GUI dropdown menu.

TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF identifies the most **distinctive** words in each document — words that are frequent in one document but rare across the rest of the corpus. This makes it fundamentally different from simple word frequency counts, which are dominated by common words.

How it works

- **TF (Term Frequency):** How often a word appears in a document (with sublinear scaling: 1 + log(tf) to dampen the effect of very frequent words).
- **IDF (Inverse Document Frequency):** log(N / df), where N = total documents, df = how many documents contain the word. Common words get low IDF; rare words get high IDF.
- **TF-IDF = TF × IDF.** A word scores high when it is frequent in one document but rare in others.

Output

- TF-IDF score matrix (csv): all words × all documents with their TF-IDF scores.
- Top 20 words per document (csv): the most distinctive words for each document.
- Bar chart (png): visual comparison of top words.
- Cosine similarity matrix (csv + heatmap png): how similar each pair of documents is based on their TF-IDF vectors (1.0 = identical vocabulary profile, 0.0 = completely different).

Requirements. Multiple documents (a directory of .txt files). TF-IDF is meaningless for a single document, since IDF requires comparing across documents.

Lexical Diversity

Lexical diversity measures how varied the vocabulary is in a text. A text that uses many different words has high lexical diversity; a text that repeats the same words has low diversity.

Five measures are computed:

1. **TTR (Type-Token Ratio)** = unique words / total words. Simple but sensitive to text length — longer texts always have lower TTR, because words inevitably repeat.
2. **Root TTR (Guiraud's Index)** = unique words / sqrt(total words). Partially corrects for text length by using the square root.
3. **Log TTR (Herdan's C)** = log(unique words) / log(total words). Another length correction, using logarithms.
4. **MTLD (Measure of Textual Lexical Diversity).** Reads through the text sequentially, tracking TTR. Each time TTR drops below 0.72 (a "factor"), it resets. MTLD = total words / number of factors, computed in both directions and averaged. Less sensitive to text length than TTR.
5. **vocd-D.** Takes random samples of different sizes (35–50 words, 100 trials each), computes TTR for each sample, then fits a mathematical curve to the TTR-versus-sample-size relationship. The D parameter of the fitted curve is the diversity score: higher D means more diverse vocabulary. The most robust to text length of all five measures.

Interpretation

- TTR values above 0.5 suggest diverse vocabulary; below 0.3 suggests repetitive text.
- MTLD values typically range from 50 to 200+ for natural text.
- vocd-D values typically range from 30 to 120+.
- Compare across documents in the same corpus for meaningful contrasts.

Word Frequency Distribution (Zipf's Law)

This tool computes the complete rank-frequency distribution of all words in the corpus and tests whether it follows Zipf's Law — the empirical observation that the frequency of a word is inversely proportional to its rank.

Zipf's Law. If the most frequent word appears 1000 times, the 2nd most frequent should appear about 500 times, the 10th about 100 times, and so on. On a log-log plot this produces a straight line with slope −1.

Output

- Word frequency table (csv): every unique word with its rank, frequency, log(rank) and log(frequency).
- Three-panel chart (png):
  - Top N words bar chart — the most frequent words.
  - Log-log rank-frequency scatter — with Zipf's ideal slope = −1 line for comparison. Natural language texts closely follow this line; deviations indicate unusual vocabulary distributions.
  - Cumulative coverage curve — how many unique words cover 50% and 90% of all tokens. Typical English text: about 100 words cover 50% of tokens; about 5000 words cover 90%.

- Summary statistics (csv): total tokens, total types, hapax legomena count (words appearing exactly once), and the 50% / 90% coverage thresholds.

Hapax legomena

Hapax legomena (words appearing exactly once) typically make up 40–60% of all unique word types in a text. A very high hapax percentage suggests diverse or specialized vocabulary; a very low percentage suggests repetitive prose.